

Cloud Data Processing and Visualization

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Synopsis

- ✓ Data wrangling is the process of transforming and structuring data from one raw form into a desired format with the intent of improving data quality and making it more consumable and useful for visualization, analytics or machine learning.
- ✓ The data wrangling process often includes transforming, cleansing, and enriching data from multiple sources. As a result of data wrangling, the data being analyzed is more accurate and meaningful, leading to better solutions, decisions, and outcomes
- ✓ Data Visualization is a graphical representation of data and it can translate information into a visual context, making things easier for the human brain to understand

Objective

- To build a cloud-based data processing and visualization system using Azure services.
- To collect, transform, and store raw data efficiently using ETL processes.
- To improve data quality through data wrangling and transformation techniques.
- To create interactive and real-time dashboards using Microsoft Power BI for better data analysis and decision-making.
- To automate workflows and notifications using Microsoft Power Automate.
- To provide scalable and secure storage using Microsoft Azure Blob Storage and Azure Data Lake.

Storage Layer



- Offers cost-effective and scalable object storage for unstructured data
- Optimized for storing massive amounts of unstructured data, such as text or binary data



- Provide easier collection and indefinite storage of all types of data. They allow companies to transform raw data into structured data suitable for SQL-based analytics, data science and machine learning, all with lower latency



- Power your resource-intensive apps and keep mission-critical workloads running at peak performance
- It offers built-in high availability and automated backups to ensure data redundancy and protection against failures

Transformation Layer

Azure Data
Factory



- It is the cloud-based ETL tool(Extract, Transform and Load)
- Provides a data integration and transformation layer that works across your digital transformation initiatives.



- Allows you to build and execute Power Query mash-ups to execute data wrangling in Data Factory & PBI.
- Graphical interface for getting data from sources and a Power Query Editor for applying transformations.

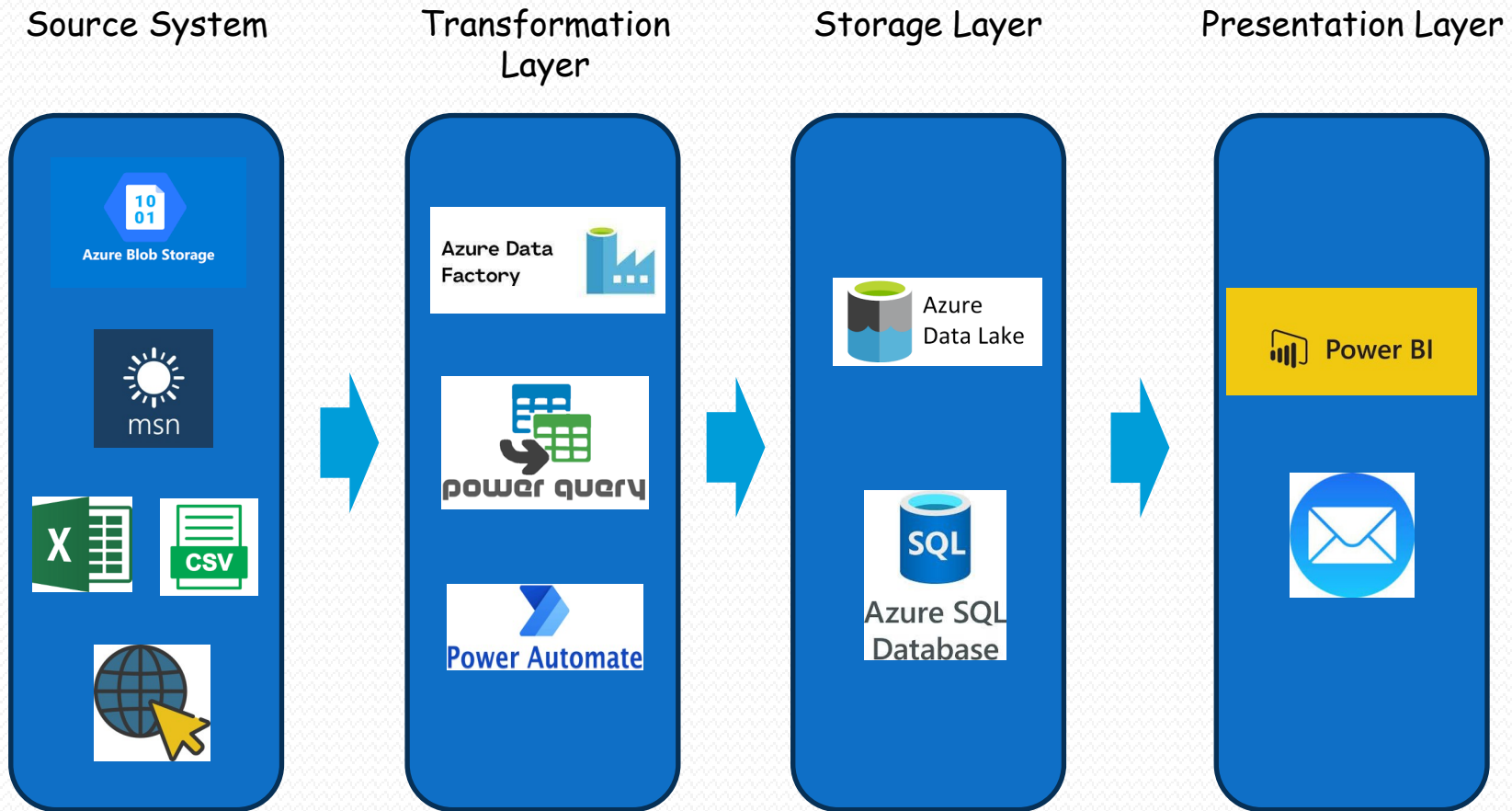


- Build custom integrations and automated processes and third-party applications without the help of developers.
- Streamline processes across organization to save time.

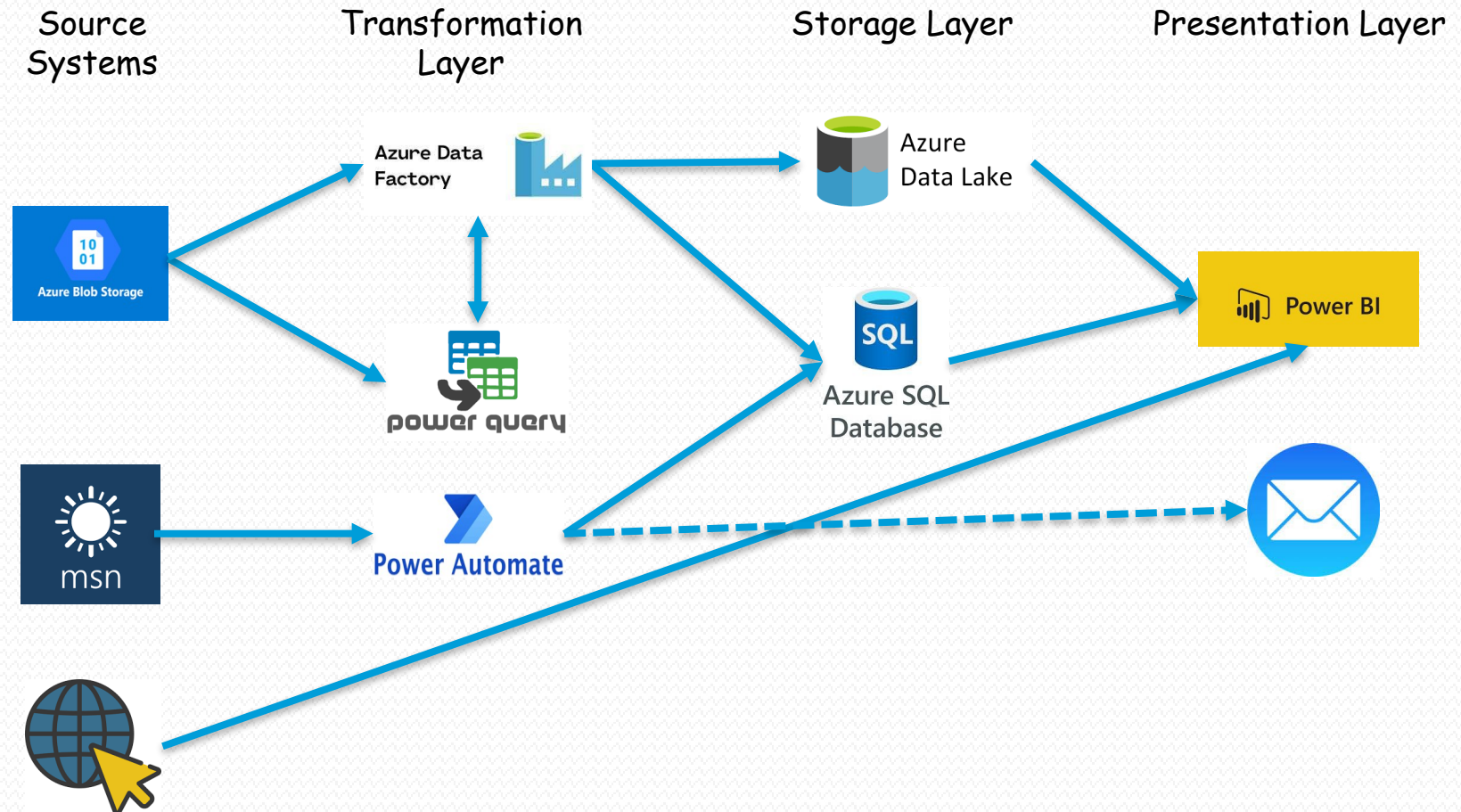
Visualization Layer

- We are using **MS-Power BI** for data visualization and the below are the key benefits
 - Customized dashboards and interactive reports
 - Seamless integration with numerous data sources and databases
 - Seamless access to your data, anywhere, on any device, at any time
 - Real-time access to information
 - Ability to customize security features
 - Economical and cost-effective
 - Multi-Dashboard deployment using Power BI Apps
 - Enables Big Data analysis and sharing business in-sight

Data-flow Diagram



Architecture Diagram



Challenges

- ✓ Handling missing values and inconsistent records in the CSV dataset.
- ✓ Transforming raw CSV data into structured and meaningful information.
- ✓ Configuring data pipelines and integrating multiple Azure services.
- ✓ Managing ETL processes and dataflows in Microsoft Azure Data Factory.
- ✓ Ensuring accurate data transformation and storage during processing.

Future Enhancement

- ✓ Integrating real-time data sources instead of static CSV files.
- ✓ Implementing machine learning models for predictive analytics.
- ✓ Adding automated real-time alerts and monitoring systems.
- ✓ Improving scalability for handling larger datasets and multiple users.

Conclusion

The Cloud Data Processing and Visualization project successfully demonstrates the use of cloud technologies for processing, storing, and visualizing CSV-based datasets. Using Microsoft Azure services, the project performs efficient data transformation and storage through ETL processes. The processed data is visualized using interactive dashboards in Microsoft Power BI, enabling better analysis and decision-making. The project highlights the practical application of cloud computing, data engineering, and business intelligence concepts in a scalable environment.



Thank you!